

Let T_n be the sum of the first n natural numbers. Then for any $n \in \mathbb{N}$,

$$T_n + T_{n+1} = (n+1)^2$$

Base case: $n=1$, then $T_n = 1$, $T_{n+1} = 3$, $(n+1)^2 = 4$

$$1 + 3 = 4 \quad \checkmark$$

Inductive Hypothesis: Suppose $T_k + T_{k+1} = (k+1)^2$ for some arbitrary $k \in \mathbb{N}$

Inductive step: Show $T_{k+1} + T_{(k+1)+1} = ((k+1)+1)^2$

By definition, $T_{k+1} = T_k + (k+1)$

$$T_{(k+1)+1} = T_{k+2} = T_{k+1} + (k+2)$$

So we have as our sum

$$(T_k + (k+1)) + (T_{k+1} + (k+2))$$

$$= \underbrace{T_k + T_{k+1}}_{(k+1)^2} + (k+1) + (k+2)$$

$$= (k+1)^2 + 2k + 3$$

$$= k^2 + 2k + 1 + 2k + 3 = k^2 + 4k + 4 = (k+2)^2 = ((k+1)+1)^2$$

$$\therefore \forall n \in \mathbb{N}, \quad 1 + \dots + n-1 + n = (n+1)^2$$

Q.E.D.